

PRINCE SAVSAVIYA

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Professional Summary

Machine Learning Engineer with strong expertise in **HPC and Distributed Systems**, specializing in building low-latency inference pipelines and optimizing high-throughput parallel workloads. Experience deploying scalable AI solutions (Kubernetes/AWS) and accelerating scientific workflows with **MPI and Distributed Computing**. Proven track record of improving search precision by 25% and reducing system latency by 40%.

Education

University of California, Riverside M.S. Computer Science, GPA 3.56/4.0	Riverside, CA	March 2026 (expected)
Adani Institute of Infrastructure Engineering, GTU B.E. Information & Communication Technology, GPA 3.6/4.0	Ahmedabad, IN	May 2024

Core Technical Skills

Machine Learning:	PyTorch, TensorFlow, Scikit-learn, BERT, LLMs (Hugging Face), Computer Vision, Physics-Informed ML
MLOps & Cloud:	Docker, Kubernetes, AWS (EKS, EC2, S3), GitHub Actions (CI/CD), Optuna, Vector Search, MLflow
Languages:	Python, C++, SQL (PostgreSQL), Bash, CUDA, C
HPC & Performance:	MPI, GPU Optimization, PETSc, DofinX, Latency Optimization
Data & Search:	Elasticsearch, Lucene, FAISS, Pandas, NumPy

Experience

Graduate Researcher — University of California, Riverside	Riverside, CA	May 2025 – Present
- Architected a modular FSI simulation framework orchestrating distributed CFD and FEM solvers via MPI, validated against Turek-Hron benchmarks for high-fidelity stability. - Optimized inter-process communication overhead in partitioned solvers by refactoring data serialization pipelines, improving multi-node simulation throughput by 2×. - Prototyping Physics-Informed ML predictors (Hamiltonian Neural Networks) to reduce fixed-point iterations; designing feature sets and data pipelines to accelerate model convergence. - Developed a parameterized benchmark generator with automated regression tests (Python/Bash), reducing experiment setup time for new physics cases by 80%.		
Research Intern — Adani University	Ahmedabad, IN	May 2023 – Dec 2023
- Built a physics-informed ML framework combining PDE ageing models with ensemble regressors (SVR, RF, GBR), cutting MSE 98% and RMSE 86% vs. data-only baseline. - Ingested 200,000+ battery-cycle records in Python; automated feature engineering and Optuna hyperparameter tuning, reducing experiment turnaround 30%. Publication: <i>Journal of Power Sources</i>, Vol. 627, 2025 (IF ≈ 7.9) — DOI 10.1016/j.jpowsour.2024.235771.		

Selected Projects

RetrieveX — Semantic Searchbo Engine	Django · BERT · FAISS · AWS	Jan 2025 – Apr 2025
- Architected a hybrid retrieval system merging BM25 keyword search with BERT+FAISS vector embeddings, improving top-3 precision 25% on 150k medical documents. - Engineered an asynchronous inference pipeline using Django and REST APIs, optimizing query latency to <200 ms (P_{95}).		
AI-Driven Infrastructure Autoscaler	LSTM · Kubernetes · Helm · CI/CD	Jan 2025 – Apr 2025
- Designed a time-series forecasting service using LSTMs to predict cluster load, automating replica scaling and cutting cloud over-provisioning costs by 18%. - Implemented MLOps best practices with automated model retraining triggers and CI/CD deployment to AWS EKS; sustained 99.9% SLA under 4× load spikes.		
Risk-Analytics Microservice	FastAPI · Pandas · Docker · AWS	Sep 2024 – Jan 2025
- Designed real-time analytics service ingesting multi-asset data with vectorized pipelines; delivered metrics in <150 ms. - Automated deployment with GitHub Actions → AWS ECR → EC2; maintained 99.9% uptime with Prometheus monitoring.		